SESSIONAL TEST-I (B.TECH 1st Sem CE)

Course Name: Physics (Introduction to Mechanics) Course Code: BPY25111T

Full Marks: 20 **Dated: 06-10-2026**

The figures in the margin indicate full marks for the questions.

1. Answer in brief. 1x5=5

(a) If 'a_0' is the linear acceleration of a frame, 'a' is the acceleration of a particle, then write the value of the pseudo force.

(b) If '$\vec{E}$' be the electric field intensity, $\vec{B}$ be the magnetic field strength, then write the value of electromagnetic force on a charged particle moving with velocity, '$\vec{v}$.

(c) Write the name of the frame of reference where Newton's laws are valid.

(d) What is the physical quantity in rotational motion that corresponds to the force in linear motion?

(e) If 'l' and 'b' be the length and breadth of a lamina, then write the value of MI about an axis passing through the centre and perpendicular to the length.

2. (a) What are inertial and non inertial frames of reference? 2

(b) Explain briefly about the fundamental forces in nature 3

(c) Write Keplar's laws. The radius of Neptune's orbit is 4.5×10^{12} m and its period of revolution round the Sun is 5.2×10^{9} sec. Find the radius of the orbit of Venus if its period of revolution is 2×10^{7} sec ? 3+3 = 6

OR

(c) Find the MI of a circular disc about an axis passing through the centre and perpendicular to its plane. 6

(d) Show that in case of bar pendulum, centre of suspension and oscillations are interchangeable. 4